

ACI Wave 1 — Foundation + Global Data Lake

Date: 2026-06-26 Status: DEPLOYED (additive)

Scope: Strictly additive. No existing tables, functions, cron jobs, or dashboards modified. New surface uses aci_* prefix.

New tables: aci_data_sources, aci_signals (admin-only RLS).

Reused tables: aci_settings, aci_runs, aci_run_steps, aci_audit_log, aci_approvals, aci_learning_events, aci_forecasts, aci_recommendations, aci_tasks.

Edge functions:

- aci-orchestrator — fans out to existing refresh functions only
- aci-data-source-health — probes 17 sources, writes freshness state

Fan-out (no new provider API calls):

```
aci-data-source-health -> prie-auto-orchestrator ->
pga-overview-sync -> pe-endpoint-matrix
```

Cron: aci-orchestrator-hourly (minute 7).

Debounce: 5 min via aci_audit_log.

Audit: every run logged to aci_runs + aci_run_steps + aci_audit_log.

Provenance rule: every aci_signals row carries source_key, captured_at, confidence — no fabricated values allowed.

Dry run (trigger=deploy):

```
aci-data-source-health    ok    2.4s
prie-auto-orchestrator   ok 129.8s
pga-overview-sync        ok    1.0s
pe-endpoint-matrix       401 (pre-existing cron-secret guard)
```

Data source snapshot:

```

healthy:   pec, prie
stale:     products
very_stale: blog_engine, gmc, orders, pinterest_pins, product_media,
            seo_engine
probes-to-refine in W2: cj, cpe, ga4, gsc, inventory, pga,
                        pinterest_ads, pinterest_analytics

```

Guardrails enforced:

- Never auto-execute financial actions
- No new provider-API write paths
- Additive only

Next: Wave 2 (Global Product AI — 12 scores per product) on confirmation.